蔡鎬駿 Chai Hao Tzun

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EDUCATION

NATIONAL CHENG KUNG UNIVERSITY 2025 - Present
M.S. in Computer Science and Information Engineering.

NATIONAL CHENG KUNG UNIVERSITY 2021 - 2025
B.S. in Computer Science and Information Engineering.

- GPA: 4.05/4.3
- Relevant Coursework: Data Structures,Algorithms, Operating Systems,Linux System & Open-Source Software, Introduction to Artificial Intelligence

EXPERIENCE

REALTEK SEMICONDUCTOR Jul - Aug 2026
Intern, System Development (SD) Team, Multimedia Business Group.

- Built an automated image-annotation pipeline that generates preliminary labels for human review, cutting annotation man-hours by over 50% while maintaining consistent label quality.
- Automated image content classification and training-data collection, eliminating manual review and letting datasets scale without added labor cost.
- Packaged the tools into a user-friendly web UI (Vite, FastAPI, Ollama) for easy handover.

PROJECTS

MentalRAG (LLM-Powered Counseling Assistant)

- Developed a RAG-based mental health advisory system in collaboration with the Ministry of Education, enabling university faculty to access professional case-based guidance for student support.
- Implemented a GitLab CI pipeline to automate Docker image builds and registry deployment.
- Led system benchmarking, using Promptfoo to evaluate response quality for high-stakes mental health scenarios.
- Optimized RAG retrieval using LangChain, achieving a 12% increase in retrieval hit rate.

EnHAR-LLM: Energy-Aware Human Activity Recognition | ACL Submission

- Developed a two-stage EnHAR-LLM framework that bridges signal restoration and semantic recognition, utilizing attention-based imputation to reconstruct coarse signals before fusing them with imputation stage representations and time-series embeddings for classification.
- Improved F1 score by 10.23% over state-of-the-art baselines and surpassed the higher-frequency (1Hz) baseline performance despite using 90% less data (0.1Hz vs 1Hz).
- Achieved a 95% relative F1 improvement under extreme downsampling (0.02Hz) using a Qwen backbone, successfully classifying minority high-movement activities such as bicycling.

SKILLS

- **Programming Languages**: Python, JavaScript, C, HTML, CSS.
- **AI & Machine Learning**: PyTorch, Hugging Face Transformers, Large Language Models (LLMs), RAG (Retrieval-Augmented Generation), LangChain, Scikit-learn.
- **DevOps & Cloud**: Docker, Google Cloud Platform (GCP), Cloud Run, CI/CD (GitHub Actions, GitLab CI), Linux, Git/Version Control, Containerization.
- **Web Development**: React.js, Next.js, TailwindCSS, RESTful APIs, Frontend, Backend.
- **Languages**: Mandarin (Native Proficiency), English (Fluent: IELTS 8.5), Malay (Fluent: SPM Grade A).

HONORS & AWARDS

- **Outstanding Student Award (NCKU)** 2021
- **Outstanding Overseas Graduate Award** 2025